Lab 2 – DBI202

Requirement Analysis

Requirement analysis is the foundational step in developing a database project. It involves understanding what the system needs to do, what data it must manage, and how that data should be structured, accessed, and maintained.

Write the requirement analysis for your chosen project including the following parts:

1. Define project objectives:
    a. What is the purpose of the database?
    b. What problems will it solve?
2. Identify stakeholders and their needs:
    a. Who are the stakeholders (e.g. users, admins, customers, …)?
    b. For each stakeholder, what operations do they need (e.g. search, update, delete which kind of data, write reports, …)?
3. Gather functional requirements: these are specific operations the system should perform.
    a. Examples:
        i. Add new records (e.g. customers, product, …)
        ii. Updating existing entries
        iii. Generate reports (e.g. monthly sales)
4. Gather non-functional requirements: these relate to performance, security, usability, etc.
    a. Examples:
        i. The database should respond to queries within 2 seconds
        ii. Only authenticated users can access sensitive data
        iii. Backup should occur daily
5. Define data requirements
    a. What are entities (tables) and their attributes
    b. What are relationships between entities?
    c. What are the data types and constraints?
    d. Examples:
        i. Student: ID, Name, DOB, Email
        ii. Course: CourseID, Name, CreditHours
        iii. Relationship: One student can enroll in many courses
6. Volume and frequency estimates
    a. How much data will be stored?
    b. How often will data be read/written/updated?
    c. Expected growth rate?